Experiment - 5

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Subject Name: System Design
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Aim:

To design a scalable real-time messenger application similar to WhatsApp or Facebook Messenger that supports one-to-one and group messaging with high availability, low latency, and reliable message delivery.

Objectives:

• Understand the architecture of a real-time messaging system.
• Identify functional requirements such as authentication, chat, and media sharing.
• Identify non-functional requirements including availability, reliability, and latency constraints.
• Analyze CAP theorem trade-offs for messaging systems.
• Design REST and WebSocket APIs for chat communication.

System Design:

Functional Requirements –

1. Client should be able to register and login into the system.
2. System should support one-to-one messaging.
3. System should support group messaging.
4. System should support sending textual as well as media messages.
5. System should preserve complete message history.
6. Client should be able to see delivered and read receipts.
7. Messages should be received in real-time.

Non-functional Requirements

- Target Scale: 1 Billion user having 100msgs per day.
- CAP Theorem – High availability >>>>>>> Consistency
- Latency : approx. 200- 300 ms .
- System should be highly reliable.

Core-Entitles of the System:

1. Users.
2. Groups.
3. Messages.

API Designs:

1. User Registration API: POST /user/register
2. User Login API: POST /user/login

One-to-One Messaging

A. Send Message (WebSocket): WS /messages/send

B. Get Chat List: GET /chat/{user_id} (Pagination Supported)

C. Get Messages of Specific Chat: GET /messages/{user_id}/{receiver_id}

Group Messaging

A. Create Group: POST /groups/create

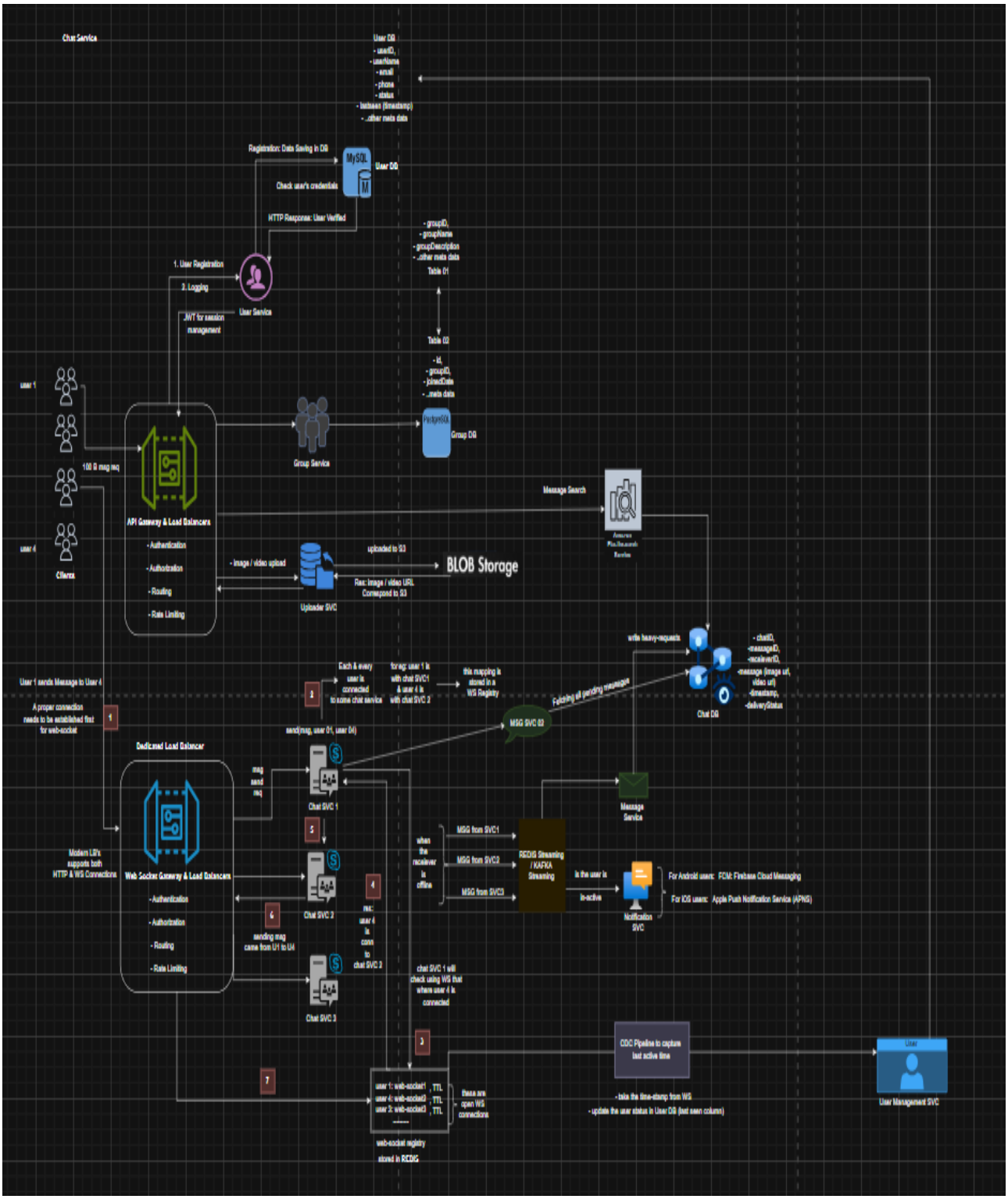
B. Send Group Message (WebSocket): WS /messages/send

C. Get Group Chats: GET /groups/{group_id} (Pagination Supported)

D. Add Member: POST /groups/{group_id}/add

E. Remove Member: DELETE /groups/{group_id}/remove

REQUIRED SYSTEM DESIGN –



Learning Outcomes :

- Designed a scalable real-time messaging system.
- Identified functional and non-functional requirements.
- Designed REST and WebSocket APIs.
- Understood high availability and eventual consistency trade-offs in messaging systems.